

Patrick Kariuki

667-770-9395 | pkariuk1@jhu.edu | [linkedin.com/in/pkariuk1](https://www.linkedin.com/in/pkariuk1) | github.com/pwkariuki

EDUCATION

Johns Hopkins University

Baltimore, MD

Bachelor of Science in Computer Science

Expected May 2026

- **Relevant Coursework:** Gateway Computing: Java, Intermediate Programming: C/C++, Data Structures, Intro Algorithms, Full-stack JavaScript, Artificial Intelligence, Computer Systems Fundamentals, Introduction to Data Science.
- **Activities:** Johns Hopkins Hackathons (24-hour GreenHacks and 36-hour HopHacks), National Society of Black Engineers (NSBE), SEO Careers Fellow, Association of Computing Machinery (ACM), ColorStack, African Students Association.

EXPERIENCE

Course Assistant for CSCI-220 Intermediate Programming

August 2024 – Present

Johns Hopkins University

Baltimore, MD

- Improved student understanding of programming techniques by mentoring students with a focus on object-oriented design in C++.
- Streamlined course management, resulting in efficient grading and exam processes, through organizing study guides, and coordinating with professors and teaching assistants.
- Fostered positive and collaborative programming experiences by holding weekly office hours to support students' learning journey.

Tech Developer Intern

May 2024 – August 2024

Sponsors for Educational Opportunity (SEO)

New York, NY

- Developed StarSight, a full-stack web application that provides real-time stargazing predictions by integrating weather data, light pollution APIs, and astronomical databases.
- Built scalable Flask backend with RESTful API endpoints, implementing JWT authentication, secure session management, and PostgreSQL database integration for user preferences and location data.
- Automated deployment pipeline using Ansible for configuration management and Docker containers, enabling consistent deployments across development and production environments.
- Collaborated within 5-person Scrum team through daily standups, biweekly sprint planning, and code reviews, delivering features on schedule across three two-week sprints.

PROJECTS

Jlox | Java, Compiler Theory

December 2024 – January 2025

- Built complete Lox programming language interpreter in Java, featuring lexical analysis, parsing, and interpretation.
- Designed a scanner and recursive descent parser to transform source code into an Abstract Syntax Tree (AST).
- Implemented visitor pattern, an object-oriented design pattern, for AST traversal and evaluation of expressions.
- Developed comprehensive error recovery strategies for both syntax and runtime errors.

Flashcard App | TypeScript, React, SQLite, Hono.js

October 2024 – December 2024

- Engineered a full-stack flashcard application with React frontend and Hono.js REST API backend, featuring user authentication, deck management, card creation, and study functionality.
- Implemented robust user authentication using Lucia Auth with secure password hashing and session management.
- Created responsive UI components using shadcn/ui and React, including interactive card flipping, pagination, search and sort capabilities.
- Designed RESTful API endpoints with comprehensive error handling, input validation using Zod, and SQLite database integration via Drizzle ORM.
- Containerized frontend and API services using Docker and automated deployment through GitHub Actions.

TECHNICAL SKILLS

Programming: Java, Python, C, C++, SQL, JavaScript/TypeScript, HTML/CSS, Kotlin, Swift, MATLAB

Frameworks: React, Node.js, Flask, Django, FastAPI

Developer Tools: Git, Docker, Ansible, Visual Studio, Android Studio, Xcode